

SOLUTIONS

Q1. Hybrid AI Recommendation Engine Design

Solution

The system combines **Content-Based Filtering (CBF)** and **Collaborative Filtering (CF)** to generate personalized recommendations. PySpark RDDs process large-scale user-item interaction data in parallel.

Architecture

User / Item Data



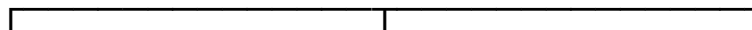
Data Ingestion

(Kafka / Files / API)



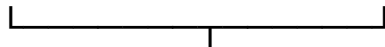
PySpark RDD Processing

(map → filter → join → reduceByKey)



Content-Based Collaborative User Behavior

Filtering Filtering Analysis



Hybrid Ranking



Intelligent Agents



Top-N Recommendations



User / App



Clicks / Likes / Ratings



Feedback Agent



Continuous Improvement

RDD Operations

- map() – transforms user/item records.
- filter() – removes invalid data.
- join() – combines user and item information.
- reduceByKey() – aggregates interactions.
- groupByKey() – groups data by user/item.
- persist() – caches frequently used RDDs.

Intelligent Agents

Profile Agent: Tracks user behavior and updates preferences.

Recommendation Agent: Combines CBF and CF results and generates Top-N recommendations.

Feedback Agent: Collects clicks, likes, and ratings and updates the recommendation process.

Conclusion

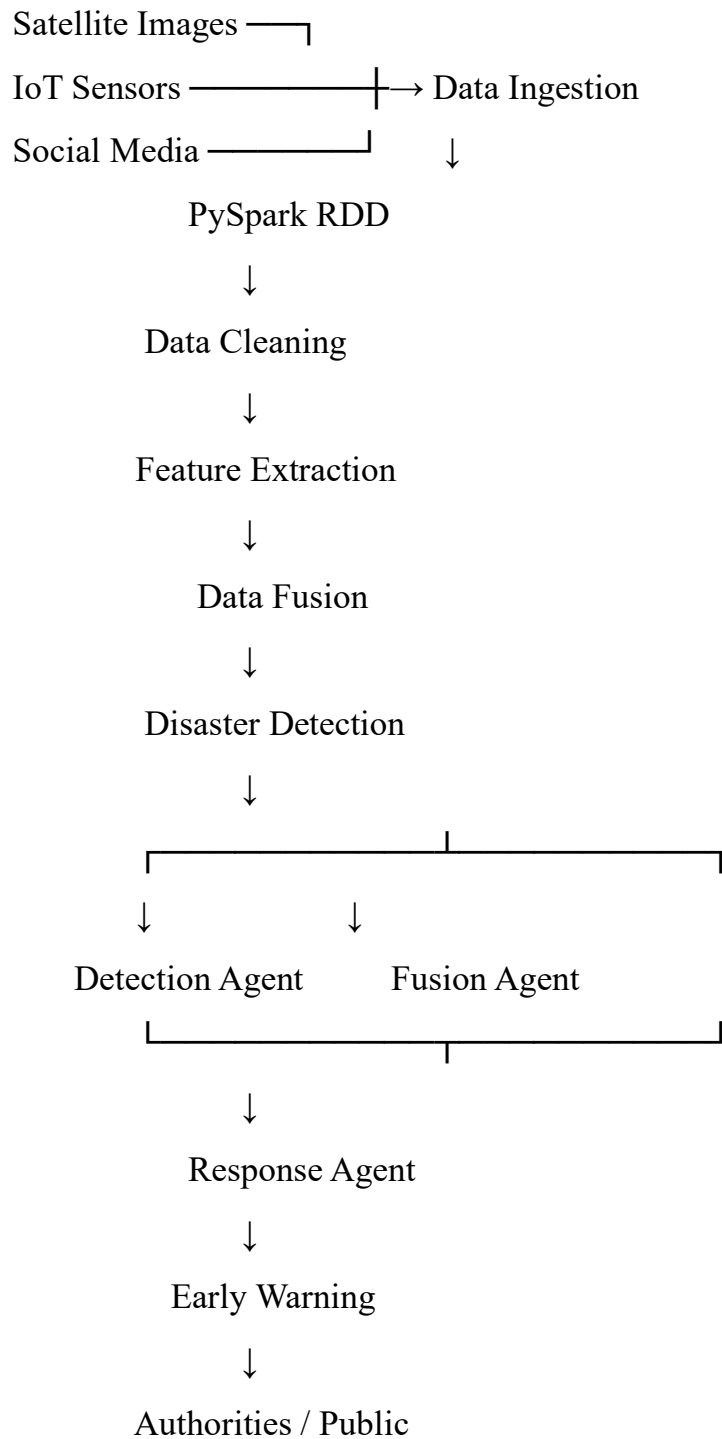
The hybrid system provides accurate and personalized recommendations while PySpark provides **distributed processing, scalability, and fault tolerance**.

Q2. Smart Disaster Detection System

Solution

The system collects information from **satellite images, IoT sensors, and social media** and processes it using PySpark RDDs. Data from different sources is fused to identify floods, earthquakes, and other disasters.

Architecture



RDD Operations

- `map()` – transforms incoming data.
- `filter()` – removes noise and invalid records.

- flatMap() – extracts multiple features.
- reduceByKey() – aggregates measurements.
- join() – combines information from different sources.
- persist() – caches important data.

Intelligent Agents

Detection Agent: Detects abnormal disaster patterns.

Fusion Agent: Combines satellite, sensor, and social-media evidence.

Response Agent: Generates early warnings and recommends response actions.

Conclusion

The system provides **fast disaster detection and early warning** by combining distributed PySpark processing with intelligent-agent coordination.

Q3. AI-Based Fraud Detection Framework

Solution

The system processes large volumes of financial transactions using PySpark RDDs and identifies suspicious transactions using anomaly detection techniques.

Architecture

Financial Transactions



Data Ingestion



PySpark RDD



Data Cleaning



Feature Engineering



Anomaly Detection



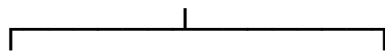
Anomaly Agent



Alert Agent



Risk Assessment



Low Risk



High Risk



Safe



Fraud Alert



Investigation



Learning Agent



Model Improvement

RDD Operations

- `map()` – parses transaction records.
- `filter()` – removes invalid transactions.
- `join()` – combines transaction and customer information.
- `groupByKey()` – groups transactions by account.
- `reduceByKey()` – calculates transaction statistics.
- `persist()` – caches frequently used data.

Intelligent Agents

Anomaly Agent: Detects unusual transaction patterns and calculates risk.

Alert Agent: Generates alerts for high-risk transactions.

Learning Agent: Learns from confirmed fraud and updates the detection model.

Conclusion

The framework provides **scalable and real-time fraud detection** and continuously improves its accuracy through feedback.

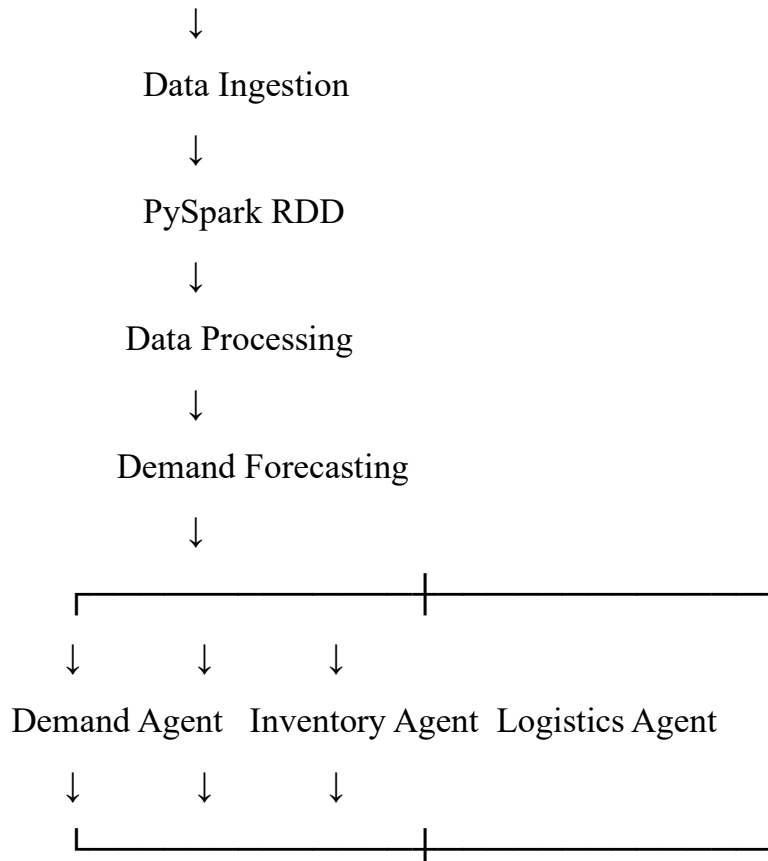
Q4. Autonomous Supply Chain Optimization System

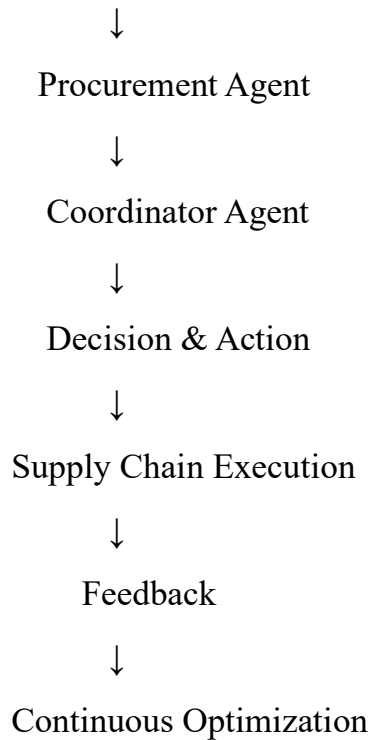
Solution

The system uses PySpark to analyze **sales, inventory, supplier, logistics, and market data**. Intelligent agents make coordinated decisions to optimize the supply chain and reduce costs.

Architecture

Sales / Inventory / Supplier / Logistics Data





RDD Operations

- `map()` – parses supply-chain records.
- `filter()` – removes incorrect records.
- `groupByKey()` – groups data by product/region.
- `reduceByKey()` – aggregates demand and costs.
- `join()` – combines sales, inventory, and logistics data.
- `persist()` – caches frequently used RDDs.

Intelligent Agents

Demand Agent: Forecasts future product demand.

Inventory Agent: Determines optimal stock and reorder levels.

Logistics Agent: Finds efficient transportation routes.

Procurement Agent: Selects suitable suppliers.

Coordinator Agent: Combines decisions from all agents.

Conclusion

The system helps achieve **cost minimization, optimal inventory, efficient logistics, and improved supply-chain performance.**

Q5. Personalized Learning Analytics Platform

Solution

The system analyzes student data such as **quiz scores, assignments, learning activities, video usage, and student interactions** using PySpark RDDs. Intelligent agents personalize the learning experience.

Architecture

Student Data



LMS / APIs



Data Ingestion



PySpark RDD Processing



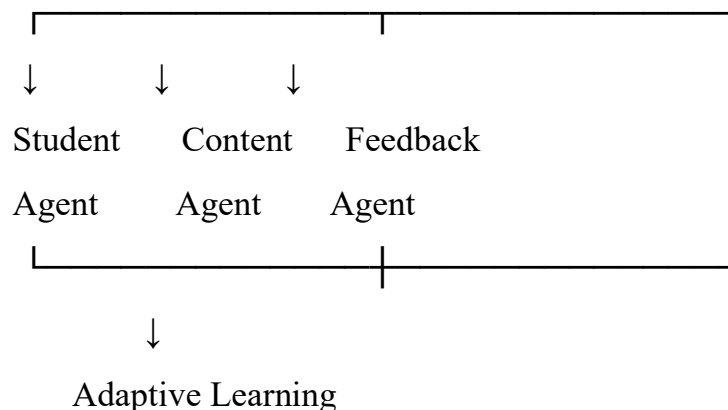
Data Cleaning

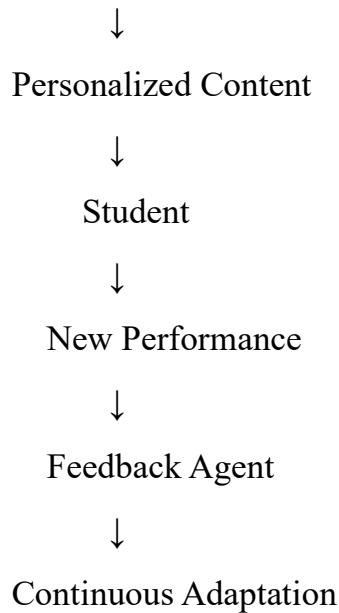


Feature Extraction



Learning Analytics





RDD Operations

- `map()` – transforms student records.
- `filter()` – removes invalid data.
- `flatMap()` – extracts learning features.
- `groupByKey()` – groups activities by student.
- `reduceByKey()` – calculates performance statistics.
- `join()` – combines student and course information.
- `persist()` – caches frequently used data.

Intelligent Agents

Student Profile Agent: Tracks student progress and maintains the learning profile.

Content Recommendation Agent: Recommends suitable learning materials and difficulty levels.

Feedback Agent: Provides real-time feedback and identifies learning gaps.

Analytics Agent: Analyzes performance and predicts learning outcomes.

Conclusion

The platform provides **personalized, adaptive, and scalable learning** by combining PySpark distributed analytics with intelligent agents and continuous student feedback.

